

C Combaind Qustion Bank BCA Sem 1

Problem Solving and Programming in C (B21CA02DC)

Combined Previous Year Question Bank

About These Notes

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These notes were created for my personal study and revision. I am sharing them here because someone else might find them useful too.

Wishing you all success in your semester exams.

Dua mein Yaad Rakhna.

Priority Markers

Use these symbols while revising to spot the most important and frequently asked questions.

★ = Asked in multiple years (High Priority)

🔥 = Very important topic (appears in different forms)

Sources:

- December 2024
 - April 2025
 - October 2025
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SECTION A

Answer any ten questions in a word or sentence each. Each question carries 1 mark.

1. _____ is a pictorial representation of an algorithm. (December 2024, October 2025) ★
2. Who developed the C language? (December 2024)
3. Hexadecimal integers are preceded with _____ (December 2024)
4. What is the priority of operators , / and % in C language? (December 2024)* 🔥
5. The library functions printf() and scanf() are associated with header file (December 2024)
6. Predict the output of the following program: (December 2024)

```
int main() {  
    int i;  
    for(i=0; i<25; i++);  
    printf("%d", i);  
    return 0;  
}
```

1. Which is the End delimiter of a string or character array? (December 2024)
2. Which operator is used to access the value stored at a pointer's address? (December 2024)
3. Define nested function. (December 2024)
4. The default parameter passing technique is ____ (December 2024, April 2025) ★🔥
5. List down any two type conversion methods. (December 2024, October 2025) ★
6. What is the default storage class in C? (December 2024, October 2025) ★🔥
7. Which function is used to open a file in C? (December 2024, October 2025) ★

8. What does argc represent in the context of command-line arguments?
(December 2024, April 2025, October 2025) ★🔥
9. Give an example of a formatted file output function. (December 2024)
10. The computer understands only _____ language. (April 2025)
11. _____ statement changes the control flow of code to another statement without any condition. (April 2025)
12. List down any two primitive datatypes in C. (April 2025)
13. What will be the value stored in variable 'a', `int a = 4.5 + 6.5;` ? (April 2025)
14. What does the format specifier '%d' represent in printf()? (April 2025)
15. What is the syntax of a for loop in C? (April 2025, October 2025) ★
16. How do you declare an array of 20 integers in C? (April 2025)
17. Which function is used to find the length of a string in C? (April 2025)
18. What is the return type of a function that does not return any value in C?
(April 2025)
19. What is recursion in C? (April 2025, December 2024, October 2025) ★🔥
20. Which method is the default in C (call by value / call by reference)? (April 2025, December 2024) ★🔥
21. How do you access a member of a structure using a structure variable?
(April 2025)
22. What is the scope of the 'extern' class specifier? (April 2025, October 2025) ★
23. Which function is used to move a file pointer to a specific position within a file? (April 2025)
24. What will be the output? (April 2025)

```
void main() {  
    register x=10;  
    printf("%d", &x);  
}
```

1. Define an algorithm. (October 2025, December 2024) ★🔥

2. What is the function of a language translator? *(October 2025)*
 3. In which section are the symbolic constants declared? *(October 2025, April 2025)* ★
 4. Which function is used to get formatted input in C? *(October 2025, December 2024)* ★
 5. Write the output of the following code: `printf("%3d", 0);` *(October 2025)*
 6. Give an example for a loop that tests condition at the bottom of the loop. *(October 2025, December 2024)* ★🔥
 7. Define a function. *(October 2025)*
 8. What is a structure data type? *(October 2025, December 2024, April 2025)* ★🔥
 9. What is the default return type of a function? *(October 2025)*
 10. Which function is present in all C programs? *(October 2025)*
 11. What is meant by life span of a variable? *(October 2025)*
 12. Which mode is used to append to a file? *(October 2025, December 2024)* ★
 13. Which function is used to read integer value from a file? *(October 2025)*
 14. Which is the directive that defines a macro? *(October 2025, April 2025)* ★🔥
 15. What is a pre-processor? *(October 2025, April 2025)* ★🔥
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SECTION B

Answer any five questions in two or three sentences each. Each question carries 2 marks.

1. Define Algorithm. *(December 2024, October 2025)* ★🔥
2. What do you mean by C tokens? List down various tokens. *(December 2024, October 2025)* ★🔥
3. Write down the usage of the conditional operator. *(December 2024, October 2025)* ★

4. Write short notes on printf() and scanf() function. (December 2024) 🔥
5. What is the significance of malloc() and calloc() functions? (December 2024, April 2025) ★🔥
6. What do you mean by user-defined functions? Write down the advantages of using function. (December 2024, October 2025) ★🔥
7. Discuss the advantages and disadvantages of structure. (December 2024, April 2025) ★
8. Define Static variable. (December 2024)
9. Compare type conversion and type casting. (December 2024, October 2025) ★🔥
10. Define command line arguments. (December 2024, October 2025, April 2025) ★🔥
11. Write a C program to print a message. (April 2025)
12. What is the difference between variables and symbolic names? (April 2025)
13. What is the purpose of putchar() and puts() functions in C? (April 2025)
14. How does a for loop differ from a while loop in C? (April 2025, October 2025) ★
15. Define one dimensional array with example. (April 2025, December 2024) ★
16. What is the difference between & and * in pointers? (April 2025) 🔥
17. Discuss any two advantages and disadvantages of Recursion. (April 2025, October 2025) ★🔥
18. Write a program to swap two numbers using call by reference. (April 2025, December 2024) ★🔥
19. Write a short note on preprocessor directive. (April 2025, October 2025) ★🔥
20. What is the significance of argc and argv in command line arguments? (April 2025, December 2024, October 2025) ★🔥
21. What are the steps involved in problem solving? (October 2025, December 2024) ★🔥

22. Explain any two tokens. *(October 2025)*
23. Explain if—else if ladder. *(October 2025, April 2025)* ★🔥
24. Write a program to print 1 to 10 using for loop. *(October 2025)*
25. Write a program to check whether a number is odd or even using user defined function. *(October 2025)*
26. Compare recursion and iteration on two points. *(October 2025, April 2025)* ★🔥
27. Write the output of the following code: *(October 2025)*

```
void main() {  
    printf("Hai");  
    main();  
    return 0;  
}
```

1. Mention the initial value of the following storage classes: a) extern b) Register *(October 2025, December 2024, April 2025)* ★🔥
2. What are command line arguments? *(October 2025)*
3. Give the use of rewind function. *(October 2025, December 2024)* ★

SECTION C

Answer any five questions in a paragraph each. Each question carries 4 marks.

1. Explain approaches in problem-solving. *(December 2024, October 2025)* ★🔥
2. What do you understand by identifiers and keywords? *(December 2024, April 2025)* ★🔥
3. Explain entry-controlled loop and exit-controlled loop. *(December 2024, October 2025)* ★🔥
4. Write short notes on the declaration and initialization of arrays. *(December 2024, April 2025)* ★

5. Differentiate call by value and call by reference with example. *(December 2024, October 2025, April 2025)* ★🔥
6. Explain various string manipulation functions. *(December 2024, April 2025)* ★
7. Explain local and global variables. *(December 2024)*
8. Compare array and structure. *(December 2024, April 2025)* ★🔥
9. Write short notes on register variable. *(December 2024, April 2025)* ★
10. List any 4 modes of opening files in C and explain. *(December 2024, October 2025)* ★🔥
11. Write an algorithm to find the sum of the first 100 natural numbers. *(April 2025)*
12. List down the rules for naming identifiers. *(April 2025)*
13. Describe constants in C. *(April 2025, October 2025)* ★
14. Write down the difference between break and continue statement. *(April 2025, October 2025)* ★🔥
15. Explain the functions malloc(), calloc(), realloc(), and free() in C with examples. *(April 2025, December 2024)* ★🔥
16. Write a program to find sum of squares of n natural numbers using do while loop. *(April 2025)*
17. Write down the differences between recursion and iteration. *(April 2025, October 2025)* ★🔥
18. Compare Structure and Union with example. *(April 2025)*
19. Which functions are used for reading and writing operations in file? Explain. *(April 2025, October 2025)* ★🔥
20. Draw a flow chart to find biggest among two numbers. *(April 2025)*
21. Explain type conversion with example. *(October 2025, December 2024)* ★🔥
22. Write a program to find the largest of three using conditional operator. *(October 2025, December 2024)* ★
23. Explain the structure of a C program. *(October 2025)*

24. Write a program to print the positive difference between two numbers. *(October 2025)*
 25. Explain switch case with an example. *(October 2025, April 2025)* ★🔥
 26. Explain the difference between break and continue with examples. *(October 2025)*
 27. What are the advantages of functions? *(October 2025, December 2024)* ★
 28. Explain call by value and call by reference. *(October 2025)*
 29. List any four basic file operations. *(October 2025)*
 30. Compare macros and functions. *(October 2025, April 2025)* ★🔥
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SECTION D

Answer any two questions in three pages each. Each question carries 15 marks.

1. Write a detailed note on the algorithm, flowchart, and symbols used in the flowchart with example. *(December 2024)* 🔥
2. Explain different loop control structures used in C programs with examples. *(December 2024, October 2025)* ★🔥
3. Explain recursion and its types with an example program. *(December 2024, April 2025, October 2025)* ★🔥
4. Write a C program to read n array elements and sort in ascending order. *(December 2024)*
5. Explain various operators used in C with example. *(April 2025, October 2025)* ★🔥
6. Discuss about if-else-if ladder and switch statement with example. *(April 2025, October 2025)* ★🔥
7. Discuss different types of arrays with examples, and write a program that demonstrates the use of a one-dimensional array to find the average of 10 numbers. *(April 2025, December 2024)* ★🔥

8. Describe storage classes with examples. (April 2025, December 2024, October 2025) ★🔥
9. Explain operators in C. (October 2025)
10. Explain control structures and looping in C. (October 2025)
11. Write a program to find the largest and second largest element in an array using function. (October 2025)
12. Write a program to implement file copy using command line arguments. (October 2025, April 2025) ★🔥

Important & Repeatedly Asked Topics

Topic	Asked In	Sections	Priority
Algorithm & Flowchart (definition, symbols, problem-solving steps, draw flowchart)	Dec 2024, Apr 2025, Oct 2025	A, B, C, D	★🔥
Operators in C (priority, types, conditional operator, examples)	Dec 2024, Apr 2025, Oct 2025	A, B, C, D	★🔥
Loops (for, while, do-while, entry/exit controlled, loop control structures)	Dec 2024, Apr 2025, Oct 2025	A, B, C, D	★🔥
Call by Value & Call by Reference (default method, swap program, differentiate)	Dec 2024, Apr 2025, Oct 2025	A, B, C, D	★🔥
Recursion (definition, types, advantages/disadvantages, vs iteration, example program)	Dec 2024, Apr 2025, Oct 2025	A, B, C, D	★🔥
Functions (user-defined, advantages, odd/even program, return type, main)	Dec 2024, Apr 2025, Oct 2025	A, B, C, D	★🔥
Arrays (declaration, 1D array, sort, largest/second largest, average program)	Dec 2024, Apr 2025, Oct 2025	A, B, C, D	★🔥
Pointers (& and *, dereferencing, pointer address)	Dec 2024, Apr 2025	A, B	★🔥
Structures & Unions (definition, advantages/disadvantages,	Dec 2024, Apr 2025, Oct 2025	A, B, C	★🔥

Topic	Asked In	Sections	Priority
compare with array, member access)			
Storage Classes (auto, extern, static, register — default, scope, initial values)	Dec 2024, Apr 2025, Oct 2025	A, B, C, D	★🔥
File Handling (fopen, modes, append, fread/fwrite, rewind, file pointer, file copy)	Dec 2024, Apr 2025, Oct 2025	A, B, C, D	★🔥
Command Line Arguments (argc, argv, significance, file copy program)	Dec 2024, Apr 2025, Oct 2025	A, B, D	★🔥
Preprocessor & Macros (directives, #define, macros vs functions)	Apr 2025, Oct 2025	A, B, C	★🔥
Dynamic Memory (malloc, calloc, realloc, free)	Dec 2024, Apr 2025	B, C	★🔥
printf() & scanf() (header file, format specifiers, short notes)	Dec 2024, Apr 2025, Oct 2025	A, B	★🔥
C Tokens (types, identifiers, keywords, constants, symbolic names)	Dec 2024, Apr 2025, Oct 2025	B, C	★🔥
Control Structures (if-else-if ladder, switch-case, break, continue, goto)	Apr 2025, Oct 2025	B, C, D	★🔥
Type Conversion & Casting (methods, compare with type casting)	Dec 2024, Oct 2025	A, B, C	★🔥
String Handling (delimiter, strlen, string manipulation functions)	Dec 2024, Apr 2025	A, C	★
Output Prediction (for loop trap, printf format, register, infinite recursion)	Dec 2024, Apr 2025, Oct 2025	A, B	🔥
Structure of C Program (sections, symbolic constants)	Oct 2025	C	★